Exact Tree–Theta-Trinet Collisions under the Kimura 2- and 3-Parameter Models

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Abstract

We construct exact three-leaf distributions shared by a tree and a binary semi-directed strict level-two theta trinet under both the Kimura two-parameter (K2P) and three-parameter (K3P) substitution models. The compact K2P construction lies in the strict stochastic interior and is an exact counterexample to Lemma 5.6 and the K2P part of Corollary 5.8 as stated in Version 2 of Brits, Holtgrefe, van Iersel, and Martin. Their Version 3 withdraws those K2P claims, identifies the leaf-order obstruction in the proposed induction, and leaves high-level K2P distinguishability open; our collision answers that question negatively. A second exact K2P witness is edgewise strictly continuous-time and is independently verified by ordinary-state Markov pruning. The fixed K2P theta map has full ambient rank 9 at both collisions; near the compact witness, its collision locus with the 6-dimensional tree model has dimension 17 and codimension 3, and a symmetric ansatz contains a smooth local six-dimensional family of positive collisions. The exact K3P construction likewise answers the Version 3 high-level K3P question negatively. Its fixed theta map has full rank 15, its local collision locus has dimension 23, and an implicit-function argument moves the collision into the edgewise strictly continuous-time K3P cone. Here "edgewise" permits edge-specific generators and rate ratios and imposes no common generator, molecular clock, or global temporal compatibility. The examples do not affect the corrected paper's JC or level-one results. Standard-library verifiers reconstruct the exact identities, inequalities, rank determinants, tangent data, and direct-pruning comparisons.

Keywords: phylogenetic networks; identifiability; Kimura substitution models; group-based models; algebraic statistics; edgewise continuous-time embeddability.

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1 Introduction

Identifiability asks whether an exact leaf-pattern distribution determines the underlying phylogenetic topology. Brits, Holtgrefe, van Iersel, and Martin establish full level-one identifiability under JC, K2P, and K3P and arbitrary-level tree–network distinguishability under JC [2]. Version 2 of their paper additionally claimed arbitrary-level K2P tree–trinet distinguishability and a corresponding global tree–network result [1]. Version 3 removes those K2P statements, explains that the proposed induction cannot preserve the leaf ordering required by its K2P invariant, and poses high-level K2P and K3P trinet distinguishability as open questions [2].

Algebraic work on group-based phylogenetic networks has established generic and, under additional hypotheses, full identifiability results for level-one models [8, 7]. Recent results determine dimensions for broad classes of level-one group-based networks and analyze the three-leaf 3-sunlet, the minimal level-one network [6, 3]. At level two, generic identifiability under JC is known for binary triangle-free strongly tree-child networks [4]. The theta topology studied here is instead a strict level-two trinet and admits no tree-child rooting. Our question is full pointwise tree–network distinguishability on that topology, rather than generic identifiability within a tree-child class.

These questions matter for inference from molecular sequence data because an exact model collision cannot be resolved by increasing sequence length alone: even the population site-pattern distribution is compatible with both topologies. Three-taxon marginals are also natural local summaries of larger networks. Consequently, determining which blob structures are detectable from trinets sets a limit on reconstruction procedures that rely only on those three-taxon distributions under the assumed substitution model [8].

We report two exact collision phenomena that resolve both newly explicit high-level Kimura questions negatively. First, the Version 2 arbitrary-level K2P statement fails on a strict level-two theta trinet. The compact witness in Section 3 lies in the strict interior of the source paper's stochastic space and is exactly a tree distribution. A separate sign point and a displayed-child audit give exact witnesses to the leaf-order obstruction now noted in Version 3. Second, the same theta topology has an exact K3P tree collision.

The collisions are not isolated numerical curiosities. Under K2P, the fixed theta map has full rank in the nine-dimensional affine K2P Fourier space, while the tree model has dimension six. Under K3P the corresponding dimensions are fifteen and nine, and the theta map again has full ambient rank. The resulting collision loci are positive-dimensional but proper in network parameter space. These calculations imply neither universal distinguishability nor generic indistinguishability: on this theta topology, tree equivalence is nongeneric but occurs along substantial exact families.

The present examples concern the removed arbitrary-level K2P trinet/global statements and answer the open high-level K2P and K3P trinet questions. They do not test the JC trinet inequality, the level-one K2P sunlet calculation, the level-one full-identifiability theorem, the combinatorial restriction lemma, K3P quartet distinguishability, or every possible restricted level-two theorem.

Main results.

- An exact K2P tree–theta collision over $\mathbb{Q}(\sqrt{71})$, with all transition probabilities positive and exact minimum pattern probability 1188799/79626240.
- A fixed-order diagnosis and an independent rational theta point for which the proposed K2P invariant is negative in all six leaf orders.
- An exact edgewise strictly continuous-time K2P collision, verified by both Fourier inversion and direct Markov pruning.
- Full K2P rank 9 at both exact collisions, a local 17-dimensional collision locus, and an exact six-dimensional symmetric collision family.
- An exact K3P collision over $\mathbb{Q}(5^{-1/4})$, full rank 15, a local 23-dimensional collision locus, and an edgewise strictly continuous-time analytic extension.

2 Models and the theta topology

Let $G = \{A, C, G, T\} \cong \mathbb{Z}_2^2$, with A the identity, $C + G = T$, and the symbol order A, C, G, T. We use the uniform stationary root distribution. If p_{ijk} is an ordinary site-pattern probability and

$\chi_x(i)$ is the character indexed by x, our Fourier normalization is

$$q_{xyz} = \sum_{i,j,k\in G} \chi_x(i)\chi_y(j)\chi_z(k)p_{ijk}, \qquad p_{ijk} = \frac{1}{64}\sum_{x,y,z\in G} \chi_x(i)\chi_y(j)\chi_z(k)q_{xyz}, \tag{1}$$

with character table (rows and columns both in A, C, G, T order)

$$(\chi_x(i))_{x,i} = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}.$$

Thus $q_{AAA} = 1$, and a three-leaf Fourier coordinate can be nonzero only when $x + y + z = A$. Sixteen of the sixty-four coordinates are consistent. The transform in (1) is invertible, so equality of all Fourier coordinates is equivalent to equality of all sixty-four ordinary site-pattern probabilities. These are the standard group-based Fourier coordinates [5, 9].

A K3P edge has Fourier vector $a = (1, a_C, a_G, a_T)$. Its transition probabilities are

$$p_A = (1 + a_C + a_G + a_T)/4, \qquad p_C = (1 + a_C - a_G - a_T)/4,$$
$$p_G = (1 - a_C + a_G - a_T)/4, \qquad p_T = (1 - a_C - a_G + a_T)/4.$$

Following both source versions, let $\Theta_0(N)$ denote the restricted parameter set in which every edge matrix is real and stochastic with entries in $[0, 1]$, is nonidentity and positive definite, and has nontrivial Fourier eigenvalues in $(0, 1)$; every inheritance parameter lies in $(0, 1)$. We write $\Theta_0^\circ(N) \subseteq \Theta_0(N)$ for the strict stochastic interior in which every transition-matrix entry is positive. All explicit collision witnesses below lie in the appropriate Θ_0° spaces. K2P is the submodel $a_C = a_T$, written $(1, s, g, s)$.

For a three-star tree with pendant vectors α, β, γ,

$$q^T_{xyz} = \alpha_x\beta_y\gamma_z \qquad (x + y + z = A).$$

At a network, each reticulation selects one parent and the distribution is the inheritance-weighted mixture of displayed trees.

The common rooted calculation topology has arcs

$$\rho \to 1,\ \rho \to u,\ u \to p,\ u \to q,\ p \to r_2,\ q \to r_2,\ p \to r_3,\ q \to r_3,\ r_2 \to 2,\ r_3 \to 3. \tag{2}$$

For the unrestricted fixed-topology map, let E_1 be the effective leaf-1 vector obtained by composing the two root-adjacent arcs, let D_i be the pendant vector on $r_i \to i$, and retain U and V for $u \to p$ and $u \to q$. Write A_i for the vector on $p \to r_i$ and B_i for the vector on $q \to r_i$. The inheritance probability of the p-parent at r_i is δ_i; that of the q-parent is $1 - \delta_i$. Reticulation choices are independent, so switching weights are products of these two probabilities.

For $x + y + z = A$, the unrestricted theta map is therefore

$$\begin{aligned} F_{xyz} = E_{1,x}D_{2,y}D_{3,z}\big[&\delta_2\delta_3 A_{2,y}A_{3,z}U_{y+z} + \delta_2(1-\delta_3)A_{2,y}B_{3,z}U_yV_z \\ &+ (1-\delta_2)\delta_3 B_{2,y}A_{3,z}V_yU_z + (1-\delta_2)(1-\delta_3)B_{2,y}B_{3,z}V_{y+z}\big], \end{aligned} \tag{3}$$

and $F_{xyz} = 0$ for inconsistent labels. The nine effective semi-directed edges give 18 edge parameters under K2P and 27 under K3P; together with (δ_2, δ_3), the fixed-topology parameter dimensions are respectively 20 and 29.

The repeated labels K, S, T are specializations used only for the explicit closed-form witnesses. For each such witness,

$$E_1 = K^{\odot 2}, \quad D_2 = D_3 = K, \quad A_2 = A_3 = S, \quad B_2 = B_3 = T, \quad \delta_2 = \delta_3 = \tfrac{1}{2}.$$

The rank minors and the implicit-function branch below use the unrestricted map (3), so the two A_i vectors, the two B_i vectors, all three pendant vectors, and both inheritance probabilities may vary independently. In rooted calculations E_1 is represented as the product of the vectors on $\rho \to 1$ and $\rho \to u$.

After suppressing the degree-two root, the maximal cyclic core consists of paths

$$p - u - q, \qquad p - r_2 - q, \qquad p - r_3 - q.$$

It has three incident leaf edges and two reticulations, hence is a strict level-two nontrivial 3-blob. Root suppression composes the two root-adjacent edge vectors coordinatewise.

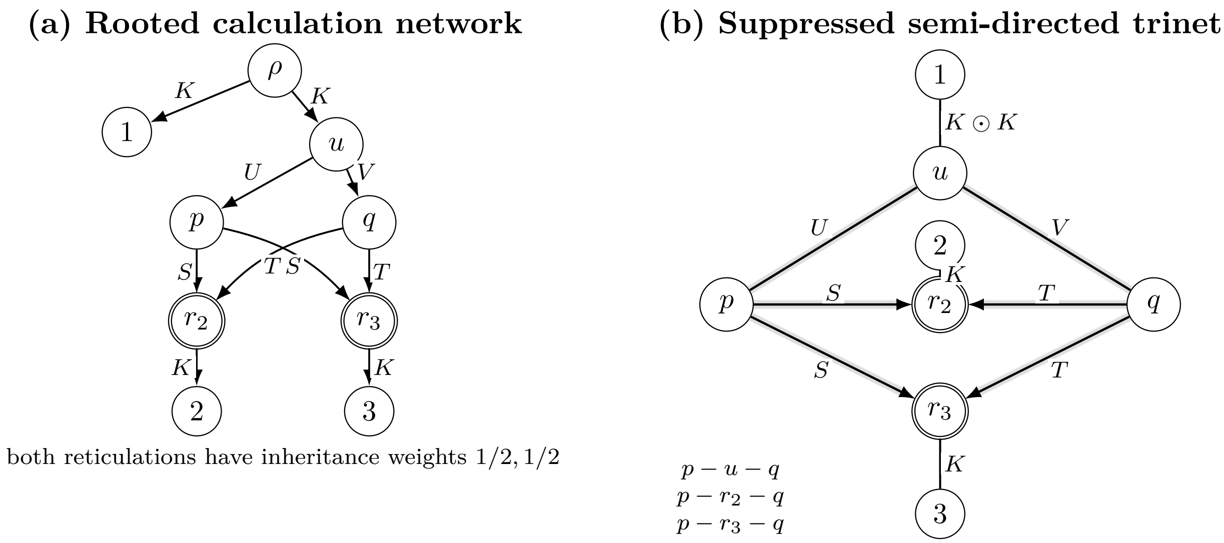

Figure 1: The rooted representation used for the symmetric collision witnesses and the suppressed semi-directed theta trinet. The repeated labels K, S, T and equal inheritance weights specialize the witnesses only; the rank and implicit-function calculations use the unrestricted edge and weight parameters in (3).

Remark 1 (Blob terminology). Under the conventions of Brits et al., the theta core is a strict level-two nontrivial 3-blob. The collision theorem does not require an additional no-2-sub-blob assumption. The accompanying topology audit records the literal incidence enumeration.

2.1 Edgewise continuous-time cones

Throughout, "edgewise strictly continuous-time" means that each edge matrix separately has the form $\exp(Q_e t_e)$, with $t_e > 0$, for a symmetric K2P or K3P generator Q_e whose relevant substitution-rate classes are all positive. The generators, rate ratios, and edge times may vary from edge to edge. We impose no common rate matrix, molecular clock, compatible node times, or other global temporal constraint.

For K3P, symmetric continuous-time rates give the following expressions on each edge, where for readability $\lambda_C, \lambda_G, \lambda_T$ denote the edge-specific products $\lambda_{C,e}t_e, \lambda_{G,e}t_e, \lambda_{T,e}t_e$:

$$a_C = \mathrm{e}^{-2(\lambda_G+\lambda_T)}, \quad a_G = \mathrm{e}^{-2(\lambda_C+\lambda_T)}, \quad a_T = \mathrm{e}^{-2(\lambda_C+\lambda_G)}.$$

All three rates are positive exactly when

$$a_C > a_G a_T, \qquad a_G > a_C a_T, \qquad a_T > a_C a_G. \tag{4}$$

For K2P, $a = (1, s, g, s)$, these reduce to

$$g > s^2. \tag{5}$$

3 An exact K2P collision over $\mathbb{Q}(\sqrt{71})$

Assign

$$\begin{aligned} K &= (1, \tfrac{1}{2}, \tfrac{1}{2}, \tfrac{1}{2}), \quad U = (1, \tfrac{4}{5}, \tfrac{19}{30}, \tfrac{4}{5}), \quad V = (1, \tfrac{7}{240}, \tfrac{239}{360}, \tfrac{7}{240}), \\ S &= (1, \tfrac{1}{4}, \tfrac{1}{2}, \tfrac{1}{4}), \quad T = (1, \tfrac{1}{3}, \tfrac{1}{27}, \tfrac{1}{3}). \end{aligned} \tag{6}$$

Use K on $\rho \to 1, \rho \to u, r_2 \to 2, r_3 \to 3$; U on $u \to p$; V on $u \to q$; S on both $p \to r_2$ and $p \to r_3$; and T on both $q \to r_2$ and $q \to r_3$.

Lemma 2 (K2P edge admissibility)**.** *The rooted network parameter lies in* $\Theta_0^\circ(N_\theta)$*, and every effective edge has strictly positive transition probabilities. The suppressed pendant edge is* $K \odot K = (1, 1/4, 1/4, 1/4)$ *with transition row* $(7/16, 3/16, 3/16, 3/16)$.

Proof. The transition rows of K, U, V, S, T are

$$(5/8, 1/8, 1/8, 1/8),\ (97/120, 11/120, 1/120, 11/120),\ (31/72, 121/1440, 289/720, 121/1440),$$

$$(1/2, 1/8, 1/4, 1/8), \qquad (23/54, 13/54, 5/54, 13/54).$$

All entries and nontrivial eigenvalues are strictly admissible. □

For $x + y + z = A$, write $x = y + z$. The four common K-edges contribute

edge	labelled descendants	Fourier factor
$\rho \to 1$	$\{1\}$	K_x
$\rho \to u$	$\{2, 3\}$	$K_{y+z} = K_x$
$r_2 \to 2$	$\{2\}$	K_y
$r_3 \to 3$	$\{3\}$	K_z

in every displayed tree. Their common product is therefore

$$K_x K_{y+z} K_y K_z = K_x^2 K_y K_z.$$

The four choices of retained incoming reticulation edges determine the remaining factors as follows.

parent at r_2	parent at r_3	desc$(u \to p)$	desc$(u \to q)$	core monomial
p	p	$\{2, 3\}$	$\varnothing$	$S_y S_z U_{y+z}$
p	q	$\{2\}$	$\{3\}$	$S_y T_z U_y V_z$
q	p	$\{3\}$	$\{2\}$	$T_y S_z U_z V_y$
q	q	$\varnothing$	$\{2, 3\}$	$T_y T_z V_{y+z}$

If a retained edge has no labelled descendant, its Fourier label is A, so its factor is $U_A = 1$ or $V_A = 1$; equivalently, that dangling branch disappears after pruning to the labelled leaves. Both inheritance probabilities are $1/2$, so each switching has weight $1/4$. After extracting the four common K-edge factors, the core mixture is thus

$$M_{y,z} = \frac{1}{4}\left(S_y S_z U_{y+z} + S_y T_z U_y V_z + T_y S_z U_z V_y + T_y T_z V_{y+z}\right). \tag{7}$$

Let $\eta = \sqrt{71}$ and set

$$P = (1, \tfrac{151}{36\eta}, \tfrac{107}{162}, \tfrac{151}{36\eta}), \qquad R = (1, \tfrac{\eta}{40}, \tfrac{31}{120}, \tfrac{\eta}{40}). \tag{8}$$

Lemma 3 (Exact K2P core factorization)**.** *For all $y, z \in G$, $M_{y,z} = P_{y+z} R_y R_z$.*

Proof. Direct substitution gives

$$M = \begin{pmatrix} 1 & 151/1440 & 3317/19440 & 151/1440 \\ 151/1440 & 71/1600 & 4681/172800 & 7597/259200 \\ 3317/19440 & 4681/172800 & 961/14400 & 4681/172800 \\ 151/1440 & 7597/259200 & 4681/172800 & 71/1600 \end{pmatrix},$$

and multiplying (8) gives the same matrix. Two useful diagnostic entries are $M_{A,C} = 151/1440$ and $M_{C,C} = 71/1600$. The certificate lists every displayed contribution. As a convention check, the same descendant-label rule reproduces all five explicit 3-sunlet Fourier coordinates displayed in Lemma 4.1 of the source paper. □

Define

$$\alpha = K^{\odot 2} \odot P = (1, \tfrac{151\eta}{10224}, \tfrac{107}{648}, \tfrac{151\eta}{10224}), \quad \beta = \gamma = K \odot R = (1, \tfrac{\eta}{80}, \tfrac{31}{240}, \tfrac{\eta}{80}). \tag{9}$$

The bound $8 < \eta < 9$ verifies their eigenvalue and transition inequalities.

Theorem 4 (Exact K2P tree–theta collision)**.** *The K2P theta distribution in (6) equals the tree distribution in (9). The network and comparison-tree parameters lie in $\Theta_0^\circ(N_\theta)$ and $\Theta_0^\circ(T)$, respectively, and both inheritance probabilities are $1/2$. The exact minimum site-pattern probability is*

$$\frac{1188799}{79626240} > 0.$$

Proof. For a consistent label,

$$q^{N_\theta}_{xyz} = K_x^2 K_y K_z M_{y,z} = (K_x^2 P_x)(K_y R_y)(K_z R_z) = \alpha_x \beta_y \gamma_z = q^T_{xyz}.$$

The forty-eight inconsistent coordinates vanish on both models. Fourier inversion gives all pattern equalities, positivity, normalization, and the minimum. □

Remark 5 (Versioned comparison with the source statements). Lemma 5.6 in Version 2 states that the displayed invariant is "zero on $\mathcal{M}_1$ and strictly positive on $\mathcal{M}_2$," and then concludes model disjointness. Its Corollary 5.8 asserts the corresponding JC/K2P disjointness when one network has a nontrivial m-blob, $m \geq 3$, and the other does not [1]. Theorem 4 is an exact counterexample to both K2P conclusions already at three leaves. Version 3 removes the K2P lemma and the K2P part of the global corollary, records the leaf-permutation obstruction, and leaves high-level K2P distinguishability open [2]; Theorem 4 answers that open question negatively.

4 A fixed-order issue in the K2P induction

The polynomial is

$$Q = q_{AGG}q_{GAG}q_{CCA}^2 - q_{AAA}q_{GGA}q_{TCG}^2. \tag{10}$$

The level-one 3-sunlet calculation remains valid when the reticulation-adjacent leaf occupies the designated final coordinate: independent substitution reproduces

$$Q = (\text{positive monomial})\,\delta(1-\delta)d_G e_G(1-f_G)^2 > 0.$$

The Version 2 arbitrary-level proof fixes the exposed leaf in that position and needs both displayed-child invariants in the same order. A child can retain a reticulation adjacent to a different leaf, whose level-one positivity is available only after another relabelling. Version 3 now identifies this ordering obstruction and withdraws the proposed K2P induction [2].

At the edgewise strictly continuous-time collision of Section 5, expose r_3. One child retains r_2. Its exact values satisfy

$$Q_{(1,2,3)} = -1.919971072382827\ldots \times 10^{-9} < 0, \qquad Q_{(1,3,2)} = 3.428488326525925\ldots \times 10^{-9} > 0.$$

Thus positivity after some child-specific relabelling does not imply positivity in the parent's fixed order. This diagnoses the proof failure; Theorem 4 independently disproves the statement.

For completeness, an independent rational theta point makes $Q < 0$ under all six leaf permutations. Write each K2P edge as the pair (s, g) corresponding to $(1, s, g, s)$ and take

	e_1	U	V	A_2	B_2
s	69/100	53/100	23/50	9/25	19/50
g	4/5	23/100	3/20	89/100	17/50

	A_3	B_3	D_2	D_3
s	9/20	3/25	17/100	49/100
g	21/100	39/50	19/100	27/100

$$\delta_2 = 3/5, \qquad \delta_3 = 11/50.$$

Here e_1 is the effective leaf-1 pendant edge, A_i is the p-parent vector, B_i is the q-parent vector, D_i is the pendant vector below r_i, and δ_i is the inheritance probability of the p-parent, consistently with (3). The network parameter lies in $\Theta_0^\circ(N_\theta)$. The six values occur in three equal pairs:

$$-\frac{16220352635472077698299088498 83}{122070312500000000000000000000000000},$$
$$-\frac{1673314326020361635172962120560 77}{19531250000000000000000000000000000000},$$
$$-\frac{510580924036090038224172288425 79}{3814697265625000000000000000000000000}.$$

Its minimum pattern probability is $2920987217429243/20000000000000000 > 0$. This point is not needed for the tree collision; it independently shows that no global fixed sign for this single ordering of Q can be used on the theta topology.

5 An edgewise strictly continuous-time K2P collision

Let ℓ be the unique root in $1.073231219980 < \ell < 1.073231219981$ of

$$634127002560\ell^3 - 2160769703472\ell^2 + 1746884136303\ell - 169873318739 = 0. \tag{11}$$

Define

$$v = \frac{73394329}{14503216}\ell^2 - \frac{1453474193}{248626560}\ell + \frac{4133719}{3669120},$$
$$x = -\frac{366971645}{99450624}\ell^2 + \frac{4259402513}{340973568}\ell - \frac{42362455}{5031936}.$$

Use the edge placement fixed in Section 2, with the new vectors K, U, V, S, T defined below:

$$K = (1, \tfrac{1}{2}, \tfrac{1}{2}, \tfrac{1}{2}),\ U = (1, \tfrac{3}{4}, v, \tfrac{3}{4}),\ V = (1, \tfrac{1}{16}, x, \tfrac{1}{16}),\ S = (1, \tfrac{1}{5}, \tfrac{1}{2}, \tfrac{1}{5}),\ T = (1, \tfrac{7}{30}, \tfrac{1}{15}, \tfrac{7}{30}). \tag{12}$$

Let $t = \sqrt{1423}$ and define

$$P_C = P_T = \frac{79}{4t}, \quad R_C = R_T = \frac{t}{240}, \quad R_G = \frac{\ell}{4},$$
$$P_G = \frac{73394329}{4762633008}\ell^2 + \frac{368713223407}{81645137280}\ell - \frac{4985775401}{1204882560}.$$

With $P_A = R_A = 1$, the four-switching matrix M from (7), evaluated at these new vectors, has the same factorization, and the tree edges are $K^{\odot 2} \odot P, K \odot R, K \odot R$.

Theorem 6 (Edgewise strictly continuous-time K2P collision)**.** *These parameters give an exact tree–theta collision. Every rooted, effective, and tree edge satisfies $g > s^2$, all transition entries are positive, the smallest edgewise continuous-time margin is* $11/900$*, and*

$$0.0149867914232177 < p_{\min} < 0.0149867914232311.$$

Proof. Exact arithmetic in the basis $1, \ell, \ell^2, t, t\ell, t\ell^2$ verifies all factorization and sign identities. Independently, ordinary-state Markov pruning on all four displayed trees agrees exactly with the comparison tree and Fourier inversion for all sixty-four patterns. □

6 K2P local rank and exact collision families

The simultaneous exchange $C \leftrightarrow T$ gives ten orbits among the sixteen consistent K2P coordinates; after $q_{AAA} = 1$, the affine K2P space has dimension nine. The positive tree model is a smooth embedded six-dimensional submanifold there. Indeed, for each $g \in \{C, G\}$ its edge eigenvalues are recovered smoothly from positive tree coordinates by

$$\alpha_g = \sqrt{\frac{q_{ggA}q_{gAg}}{q_{Agg}}}, \qquad \beta_g = \sqrt{\frac{q_{ggA}q_{Agg}}{q_{gAg}}}, \qquad \gamma_g = \sqrt{\frac{q_{gAg}q_{Agg}}{q_{ggA}}}. \tag{13}$$

Thus the positive parameterization is locally injective with a smooth inverse onto its image; in particular, its rank is six.

Proposition 7 (Full K2P rank at both collisions)**.** *At the simple collision, a selected* 9×9 *theta minor has determinant*

$$-\frac{7^2\, 11^2\, 19\, 107\, 151^2\, 15013}{2^{60}3^{25}5^{10}} \neq 0. \tag{14}$$

The same minor at the edgewise strictly continuous-time witness lies rigorously between $-4.129735 \times 10^{-22}$ *and* $-4.129729 \times 10^{-22}$.

Proof. For a K2P edge e, write e_C for its common C/T eigenvalue and e_G for its G eigenvalue. Differentiate the polynomial map (3) at the witness, using the rooted factorization of E_1 and holding the $\rho \to u$ factor fixed. The certified minor has ordered output rows

$$(F_{ACC}, F_{AGG}, F_{CAC}, F_{CCA}, F_{CGT}, F_{CTG}, F_{GAG}, F_{GCT}, F_{GGA})$$

and ordered columns

$$(a_C^{\rho\to 1}, a_G^{\rho\to 1}, U_C, U_G, V_C, V_G, (A_2)_C, (A_2)_G, (B_2)_C).$$

Exact differentiation followed by fraction-free elimination gives (14); substitution of the simple parameters also factors it as displayed there. Repeating the same symbolic differentiation over $\mathbb{Q}(\ell)$ at the edgewise strictly continuous-time witness and reducing by (11) gives the stated isolating interval. The rooted-factor derivative columns and the effective-E_1 derivative columns differ only by nonzero diagonal factors from the fixed vector on $\rho \to u$. Hence the minor also proves rank nine for the unrestricted semi-directed map. □

The theta map is therefore dominant. The semi-directed theta has twenty parameters, while the tree germ has codimension three in the ambient space.

Corollary 8 (K2P local collision geometry)**.** *Near the simple witness, theta parameters whose output lies in the tree model form a smooth* 17*-dimensional submanifold of codimension three. Generic theta parameters are not tree-equivalent, but full pointwise distinguishability fails.*

Proof. By Proposition 7, the unrestricted 20-parameter theta map is a submersion at the witness. The positive tree image is an embedded submanifold of codimension $9 - 6 = 3$ by (13). The transverse-preimage theorem therefore gives a smooth preimage of dimension $20 - 3 = 17$. □

There is also an exact family. Write

$$U = (1, u, v, u),\ V = (1, w, x, w),\ S = (1, a, b, a),\ T = (1, c, d, c).$$

The relevant entries are

$$\begin{aligned} M_{AC} &= (au + cw)/2, & M_{AG} &= (bv + dx)/2, \\ M_{CC} &= (a^2 + 2acuw + c^2)/4, & M_{GG} &= (b^2 + 2bdvx + d^2)/4, \\ M_{CG} &= (abu + adux + bcvw + cdw)/4, & M_{CT} &= (a^2v + 2acuw + c^2x)/4. \end{aligned}$$

In the positive chamber, tree factorization is equivalent to

$$M_{CG}^2 - M_{AC}^2 M_{GG} = 0, \qquad M_{CT}^2 M_{GG} - M_{AG}^2 M_{CC} = 0. \tag{15}$$

Indeed, positivity selects the unique recovery

$$R_C = \sqrt{M_{CC}}, \qquad R_G = \sqrt{M_{GG}}, \qquad P_C = M_{AC}/R_C, \qquad P_G = M_{AG}/R_G;$$

the two equations in (15) are exactly the remaining CG and CT factorization conditions, with no sign ambiguity. At the simple witness the Jacobian with respect to (v, x) is

$$675554683609333/1949951168033587200000000 > 0.$$

Corollary 9 (Exact symmetric collision family)**.** *The simple K2P witness lies in a smooth local six-dimensional semialgebraic family of positive core collision parameters satisfying* (15)*.*

After shrinking the neighborhood if necessary, openness of all strict eigenvalue, transition-probability, and inheritance inequalities keeps the network and uniquely recovered tree parameters inside their respective Θ_0° spaces.

7 An exact K3P collision

Let $h = 5^{-1/4}$. Use the same explicit edge placement fixed in Section 2. With M defined by the four-switching formula in (7), substitute the following K3P vectors:

$$\begin{aligned} K &= (1, \tfrac{1}{2}, \tfrac{1}{2}, \tfrac{1}{2}), \qquad U = (1, \tfrac{h}{3}, h, \tfrac{1}{3}), \qquad V = (1, h, \tfrac{h}{3}, \tfrac{1}{3}), \\ S &= (1, \tfrac{3h^2}{4}, \tfrac{1}{4}, \tfrac{3}{10}), \quad T = (1, \tfrac{1}{4}, \tfrac{3h^2}{4}, \tfrac{3}{10}). \end{aligned} \tag{16}$$

All inverse-Fourier rows are strictly positive. Define

$$B = (1, \tfrac{h^2}{2}, \tfrac{h^2}{2}, \tfrac{h^2}{2}), \qquad P = (1, \tfrac{5h^3+h}{4}, \tfrac{5h^3+h}{4}, h^2). \tag{17}$$

Exact reduction modulo $5h^4 - 1$ shows that this same four-switching mixture, now evaluated at the K3P vectors, satisfies $M_{y,z} = P_{y+z} B_y B_z$. The comparison tree has

$$\alpha = (1, \tfrac{5h^3+h}{16}, \tfrac{5h^3+h}{16}, \tfrac{h^2}{4}), \qquad \beta = \gamma = (1, \tfrac{h^2}{4}, \tfrac{h^2}{4}, \tfrac{h^2}{4}). \tag{18}$$

Theorem 10 (Exact K3P tree–theta collision)**.** *The theta distribution defined by* (16) *equals the tree distribution in* (18)*. The network and comparison-tree parameters lie in* $\Theta_0^\circ(N_\theta)$ *and* $\Theta_0^\circ(T)$*, respectively. This gives a negative answer to the high-level K3P trinet question posed in [2].*

Proof. The factorization (17) gives $q_{xyz}^{N_\theta} = K_x^2 K_y K_z M_{y,z} = \alpha_x \beta_y \gamma_z$ on the sixteen consistent labels; all others vanish. Exact inversion gives all pattern equalities. $\square$

The quartic point follows from the positive ansatz

$$U = (1, b\tau, b, \tau),\ V = (1, b, b\tau, \tau),\ S = (1, \kappa d, d, e),\ T = (1, d, \kappa d, e).$$

On the slice $\tau = 1/3$, the factorization equations force

$$\kappa = 3/\sqrt{5}, \qquad b^2 = 1/\sqrt{5}, \qquad e/d = 6/5,$$

so the positive branch has $b = 5^{-1/4}$.

8 K3P local overlap

The affine consistent K3P space with $q_{AAA} = 1$ has dimension fifteen. Define the ordered output and parameter lists

$$\begin{aligned} \mathcal{R} = (&\mathsf{ACC}, \mathsf{AGG}, \mathsf{ATT}, \mathsf{CAC}, \mathsf{CCA}, \mathsf{CGT}, \mathsf{CTG}, \mathsf{GAG}, \\ &\mathsf{GCT}, \mathsf{GGA}, \mathsf{GTC}, \mathsf{TAT}, \mathsf{TCG}, \mathsf{TGC}, \mathsf{TTA}), \\ \mathcal{C} = (&a_C^{\rho\to 1}, a_G^{\rho\to 1}, a_T^{\rho\to 1}, U_G, (A_2)_C, (A_2)_G, (B_2)_C, (B_2)_G, \\ &(A_3)_C, (A_3)_G, (B_3)_C, (B_3)_G, (D_2)_T, (D_3)_T, \delta_3). \end{aligned}$$

Here δ_3 is the p-parent inheritance probability at r_3, as fixed in Section 2. For the first three columns, the rooted and effective coordinates satisfy $(E_1)_g = K_g a_g^{\rho\to 1}$ for $g \in \{C, G, T\}$ because the vector on $\rho \to u$ is held fixed at K. Let

$$J_* = \left(\frac{\partial F_r}{\partial c} \right)_{r\in\mathcal{R},\, c\in\mathcal{C}} \tag{19}$$

in those row and column orders, evaluated at the K3P witness. This definition, together with the polynomial formula (3), specifies all 225 entries of the minor.

Proposition 11 (Full K3P rank and local collision geometry)**.** *At the exact collision,*

$$\boxed{\det J_* = \frac{h(10h^2+1)}{2^{61}3^4 5^{14}} > 0.} \tag{20}$$

Hence the fixed theta image contains an ordinary open neighborhood in the full fifteen-dimensional affine space. The tree model has rank nine. In the twenty-nine-dimensional semi-directed theta parameter space, the local collision locus is smooth of dimension 23 *and codimension six.*

Proof. Exact differentiation of (3) in the orders (19), followed by fraction-free Bareiss elimination in $\mathbb{Q}(h)$ with $5h^4 = 1$, gives (20); $2/3 < h < 7/10$ proves its sign. As in the K2P calculation, the fixed positive vector on $\rho \to u$ makes passage from the first rooted factor to the effective E_1 coordinates an invertible diagonal coordinate change. Thus the unrestricted fixed-topology map is a submersion.

For the positive K3P tree model, the three recovery formulas in (13) hold independently for every $g \in \{C, G, T\}$. They give a smooth local inverse and show that the tree image is a nine-dimensional embedded submanifold, of codimension six in the ambient space. The transverse-preimage theorem then gives dimension $29 - 6 = 23$. □

This does not mean generic theta parameters map to trees: the collision locus has codimension six.

9 Edgewise strictly continuous-time K3P extension

The closed-form witness satisfies two boundary equalities,

$$U_C = U_G U_T, \qquad V_G = V_C V_T,$$

with all other inequalities in (4) strict. Let θ_0 denote this witness and p_0 its ordered tuple of $\mathcal{C}$-coordinates. Set $U_C(\varepsilon) = U_C + \varepsilon$ and $V_G(\varepsilon) = V_G + \varepsilon$. Use the ordered tuple $\mathcal{C}$ as the pivot variables $p(\varepsilon)$, with $p(0) = p_0$ and every unlisted coordinate held at its θ_0 value, and impose

$$(F_r(p(\varepsilon), U_C(\varepsilon), V_G(\varepsilon)))_{r\in\mathcal{R}} = (F_r(\theta_0))_{r\in\mathcal{R}}.$$

Since the derivative with respect to p is J_*, Proposition 11 gives a unique real-analytic solution near $\varepsilon = 0$. The exact pivot tangent, in the column order $\mathcal{C}$, is

pivot parameter	derivative at $\varepsilon = 0$
$a_C^{\rho\to 1}$	$-\frac{3}{19}h - \frac{375}{304}h^3$
$a_G^{\rho\to 1}$	$-\frac{621}{152}h + \frac{1875}{304}h^3$
$a_T^{\rho\to 1}$	0
U_G	$-\frac{6}{19} + \frac{60}{19}h^2$
$(A_2)_C$	$-\frac{117}{304}h + \frac{459}{608}h^3$
$(A_2)_G$	$-\frac{75}{608}h - \frac{195}{304}h^3$
$(B_2)_C$	$-\frac{255}{608}h + \frac{135}{304}h^3$
$(B_2)_G$	$\frac{9}{304}h - \frac{9}{608}h^3$
$(A_3)_C$	$-\frac{117}{304}h + \frac{459}{608}h^3$
$(A_3)_G$	$-\frac{75}{608}h - \frac{195}{304}h^3$
$(B_3)_C$	$-\frac{255}{608}h + \frac{135}{304}h^3$
$(B_3)_G$	$\frac{9}{304}h - \frac{9}{608}h^3$
$(D_2)_T$	0
$(D_3)_T$	0
δ_3	0

Writing F_{U_C} and F_{V_G} for the corresponding two free-variable derivative columns of the ordered output map, direct substitution verifies the exact linearized identity

$$J_* p'(0) + F_{U_C} + F_{V_G} = 0.$$

The two saturated margins have derivatives

$$\frac{d}{d\varepsilon}(U_C - U_G U_T)\Big|_0 = \frac{21 - 20h^2}{19} = \frac{10h^2 - 1}{1 + 10h^2} > 0, \qquad \frac{d}{d\varepsilon}(V_G - V_C V_T)\Big|_0 = 1 > 0.$$

All other network inequalities persist by continuity. The fixed comparison tree is already edgewise strictly continuous-time: for $\beta = \gamma$ its three margins equal $h^2/4 - 1/80 > 0$, while for α, writing $a = (5h^3 + h)/16$ and $t = h^2/4$, the margins are $a(1-t), a(1-t)$, and $t - a^2 = h^2(58 - 10h^2)/256$, all positive.

Corollary 12 (Edgewise strictly continuous-time K3P collision)**.** *The same fixed, edgewise strictly continuous-time tree distribution has a nearby preimage on the theta topology for which every network edge has all three instantaneous K3P substitution-rate classes strictly positive.*

The exact certificate verifies the analytic theorem's algebraic hypotheses and tangent data; the existence of the branch is the conclusion of the implicit-function theorem.

10 Scope, implications, and open questions

The level-one K2P 3-sunlet calculation in Lemma 4.1 of [2] is not contradicted by these examples. The new conclusion is narrower: universal three-leaf tree–trinet distinguishability fails on the strict level-two theta topology. The theta trinet admits no tree-child rooting. Indeed, the directions of the four reticulation edges are fixed. In a compatible rooting, the two core edges incident with u cannot both be directed into u, so the path $p - u - q$ can point away from at most one of p and q; at least one of those tree vertices therefore has only reticulation children. Consequently, the examples do not contradict results restricted to tree-child or strongly tree-child networks, including the generic level-two JC result of [4].

For completeness, the nongenericity statement has a global algebraic justification. Let F be the complexified fixed-theta polynomial parameterization and let Y be the corresponding tree variety in the affine Fourier space. The certified full-rank minor makes F dominant. Since $\dim Y$ is strictly smaller than the ambient dimension, Y is a proper closed subvariety; hence $F^{-1}(Y)$ is a proper Zariski-closed subset of theta parameter space. The physical positive chamber is Euclidean open and Zariski dense, so generic physical theta parameters are not tree-equivalent. The dimensions 17 and 23 are the stronger local smooth-dimension statements in the physical positive chambers at the explicit witnesses. In particular, under K2P the positive collision set is locally 17-dimensional rather than isolated or confined to the stochastic boundary.

For K3P, the level-one result is likewise not addressed. At strict level two, the fixed theta map is locally surjective onto the fifteen-dimensional ambient Fourier space, and the tree-equivalent subset maps onto a neighborhood relative to the nine-dimensional tree model. Thus under both Kimura models a nontrivial three-way blob is not universally detectable from a three-leaf distribution. In biological terms, even an exact three-taxon site-pattern distribution—the infinite-data limit under the model—need not reveal whether the local ancestry contains this level-two reticulation structure. No JC collision or analogous proof failure is established here.

The work does not decide whether a useful replacement statement holds with four selected leaves, restricted blob topology, generic rather than pointwise parameters, a vector of several leaf-order invariants, or an equivalence relation generated by local theta moves.

11 Future classification problems

1. Classify the complete K2P and K3P tree–theta intersections and positive chambers.
2. Determine whether the two mechanisms belong to one symmetry-breaking family.
3. Identify local observational-equivalence moves and whether they compose independently across blobs.
4. Decide K3P quartet distinguishability and corrected higher-level K2P/K3P identifiability under additional hypotheses.
5. Determine whether JC genuinely excludes theta/tree collisions or requires another topology.
6. Characterize common-generator, clock, global-time, or edge-sharing restrictions that restore full distinguishability.

12 Exact verification and provenance

The short command `python3 verify_k2p_simple.py` checks the compact witness with the standard library. For that simple K2P witness, `python3 verify_k2p_displayed_trees.py` both derives the four Fourier monomials from the retained-edge graphs and independently performs ordinary-state Markov pruning on all four displayed trees. The complete command `python3 verify.py` reconstructs source conventions, both K2P collisions, the induction-order audit, direct Markov pruning, both K2P rank calculations, the exact family, the K3P collision, rank-15 determinant, and analytic-IFT tangent data. It ends with `ALL EXACT CHECKS PASSED`.

Code and data availability. This manuscript has no empirical data set. Its exact certificates, standard-library verifiers, source files, and replay instructions are publicly available at

https://github.com/AlecKriebel/Math/tree/k2p-k3p-theta-v1.1.0/k2p_k3p_theta_trinet_collision/k2p_k3p_theta_clarified.

The verifier differentiates the explicitly defined map (3), reconstructs the stated minors and tangent vector, checks exact field identities and isolating intervals, and independently compares ordinary-state pruning with Fourier inversion. The accompanying replay archive records the exact repository commit and SHA-256 hashes so the replay object is immutable.

Provenance and AI-assisted methods. AI-assisted analysis was used for exploratory symbolic and numerical search, code assistance, adversarial checking, and editorial review. It first identified a possible sign-definiteness and leaf-order issue in the K2P invariant and produced a high-precision collision candidate. The exact $\mathbb{Q}(\sqrt{71})$ witness, the edgewise strictly continuous-time algebraic witness, the rank calculations, and the collision-family results were then constructed and checked by exact, replayable computations. The certificates replay every computationally derived identity, sign, determinant, interval, and pruning comparison; the local analytic conclusions additionally use the stated preimage and implicit-function theorems.

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